

[Assignment-2]

Name: Imtiaz Hossain

ID : 23101137

Course : CE428
code

Section: 4

Faculty : [mom]

Initial

[Lecture-5]

[Answer to the question no 1]

a) Input image shape : 4×4
 $\uparrow \quad \uparrow$
 $m \times n$

Kernel shape : 3×3
 $\uparrow \quad \uparrow$
 $m \times n$

stride : $s = 1$

Output image shape : $H \times W$
 $\downarrow \quad \downarrow$
 4×4

we know that,

$$H = \text{floor} \left(\frac{m + 2P - m}{s} + 1 \right)$$

$$\Rightarrow 4 = \text{floor} \left(\frac{4 + 2P - 3}{1} + 1 \right)$$

$$\therefore P = 1$$

So, minimum padding width, $P = 1$

(Ans)

b)

we know that,

$$H = \text{floor} \left(\frac{m + 2P - m}{S} + 1 \right)$$

$$= \text{floor} \left(\frac{4 + 0 - 3}{1} + 1 \right)$$

$$= 2$$

[P=0 ; no padding]

$$W = \text{floor} \left(\frac{M + 2P - m}{S} + 1 \right)$$

$$= \text{floor} \left(\frac{4 + 0 - 3}{1} + 1 \right)$$

$$= 2$$

Output Image Shape: 2 x 2

Input Image:

$$\begin{bmatrix} 6 & 2 & 1 & 9 \\ 2 & 0 & 2 & 4 \\ 1 & 9 & 9 & 5 \\ 2 & 0 & 1 & 2 \end{bmatrix}$$

Gaussian Kernel:

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

① Top-left window

$$\begin{bmatrix} 6 & 2 & 1 \\ 2 & 0 & 2 \\ 1 & 9 & 9 \end{bmatrix} \times \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$B_{11} = \frac{1}{16} (6 \times 1 + 2 \times 2 + 1 \times 1 + 2 \times 2 + 0 \times 4 + 2 \times 2 + 1 \times 1 + 9 \times 2 + 9 \times 1)$$

$$\approx 2.94$$

ii) Top-right window

$$\begin{bmatrix} 2 & 1 & 9 \\ 0 & 2 & 4 \\ 9 & 9 & 5 \end{bmatrix} \times \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$B_{12} = \frac{1}{16} (2 \times 1 + 1 \times 2 + 1 \times 9 + 0 + 2 \times 9 + 4 \times 2 + 9 \times 1 + 9 \times 2 + 5 \times 1)$$

$$= \frac{1}{16} (2 + 2 + 9 + 0 + 8 + 8 + 9 + 18 + 5)$$

$$\approx 3.81$$

iii) Bottom-left window

$$\begin{bmatrix} 2 & 0 & 2 \\ 1 & 9 & 9 \\ 2 & 0 & 1 \end{bmatrix} \times \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$B_{21} = \frac{1}{16} (2 \times 1 + 0 \times 2 + 2 \times 1 + 1 \times 2 + 9 \times 4 + 9 \times 2 + 2 \times 1 + 0 \times 2 + 1 \times 1)$$

$$\approx 3.94$$

iv) Bottom-Right window

$$\begin{bmatrix} 0 & 2 & 4 \\ 9 & 9 & 5 \\ 0 & 1 & 2 \end{bmatrix} \times \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$B_{22} = \frac{1}{16} (0 \times 1 + 2 \times 2 + 4 \times 1 + 9 \times 2 + 9 \times 4 + 5 \times 2 + 0 \times 1 + 1 \times 2 + 2 \times 1)$$

$$= 4.75$$

So,

$$B = \begin{bmatrix} 2.94 & 3.81 \\ 3.94 & 4.75 \end{bmatrix}$$

After ceiling

$$\Rightarrow B = \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$$

(Ans)

(Ans)

e)

Unsharp mask, $M = \text{Original Image} - \text{Blurred}$

Here,

we take corresponding 2×2 region from input since convolution without padding reduced size.

$$I_{\text{center}} = \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix}$$

So, $M = \begin{bmatrix} 0 - \overset{3}{\cancel{2.94}} & 2 - \overset{4}{\cancel{3.81}} \\ 9 - \overset{4}{\cancel{8.94}} & 9 - \overset{5}{\cancel{4.25}} \end{bmatrix}$

$g(x,y)$ mask

$= \begin{bmatrix} -2.94 & -1.81 \\ 5.06 & 4.25 \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 5 & 4 \end{bmatrix}$

$\downarrow$

~~(Ans)~~ $= \begin{bmatrix} 0 & 0 \\ 5 & 4 \end{bmatrix}$ [After clipping]

(Ans)

2) Sharpened image using unsharp masking
($K=1$)

$g(x,y) \rightarrow$ Sharpened Image, $S = \overset{f(x,y)}{\text{Original}} + K \times$
 $= I_{\text{center}} + K \times M$ (Original - Blurred)

$= \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix} + 1 \times \begin{bmatrix} -2.94 & -1.81 \\ 5.06 & 4.25 \end{bmatrix}$

$= \begin{bmatrix} -2.94 & 0.19 \\ 14.06 & 13.25 \end{bmatrix}$

After clipping to ~~non~~ negatives into 0
 and assume max 9 (since original g_{max}).

$S = \begin{bmatrix} 0 & 0.19 \\ 9 & 9 \end{bmatrix}$

(Ans)

e) High-boost filtering ($\kappa=2$)

$$S_{HB} = I_{center} + \kappa m$$

$$= \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix} + 2 \times \begin{bmatrix} -2.99 & -1.81 \\ 5.66 & 4.25 \end{bmatrix}$$

$$= \begin{bmatrix} -5.88 & -1.62 \\ 19.12 & 17.5 \end{bmatrix}$$

clip $\rightarrow [0, 9]$

so,

$$S_{HB}^{\$} = \begin{bmatrix} 0 & 0 \\ 9 & 9 \end{bmatrix}$$

(Ans)

[Answer to the question no 2]

The giraffes are characterized by long, tall necks, and the dominant edges in this image are vertical. To detect vertical edges, we need a vertical edge detector.

A standard 3×3 Prewitt filter for vertical edges would be:

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Also, a Sobel filter (which places more weight on the center pixel) is also correct:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

also we can use 3×3 Prewitt filter for horizontal edges for the neck:

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

[Ans to the question no 3]

The eagle is flying with its wings spread wide. The dominant structure and edges of the bird are horizontal. To detect horizontal edges, we need a horizontal edge detector.

A standard 3×3 Prewitt filter for horizontal edges would be:

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Alternatively, a Sobel filter for horizontal edges:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

[Answer to the question no 4]

a)

Salt and pepper Noise

↳ The image contains random scattering of black pixels ("pepper") and white pixels ("salt") over the image. This is characteristic of impulse noise, where random pixels are set to the minimum (0) or maximum (255) intensity values.

b)

Median Filter

↳ A median filter is a non-linear filter that replaces each pixel's value with the median value of the neighboring pixels.

Salt and pepper noise consists of outliers (extreme values ~~into~~ of 0 or 255). Linear filters (like a box blur or mean filter) would simply average these extreme values into the surrounding pixels, blurring the noise rather than removing it. A median filter effectively ignores the outliers because the median

of a neighborhood is almost always a valid pixel from the background, completely eliminating the 'salt' and 'pepper' spots while preserving sharp edges better than a blurring filter.

c) Laplacian Kernel

Figure 3 shows the "soft edges" of the image with a gray background.

This implies that regions of constant intensity have been zeroed out (resulting in gray), while areas of rapid change (edges) are highlighted. This ~~the~~ is the output of a second-derivative edge detection filter, typically a Laplacian filter.

A standard 3×3 Laplacian kernel that produces this result is:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Q

Yes, figure 3 can be used to sharpen figure 2 by Laplacian sharpening.

Steps:

- 1) Figure-3 represents the high-frequency components (edges / details) of the image.
- 2) Add the edge image (figure 3) back to the original clean image (figure 2).
- 3) By adding the edges back to the original, we can amplify the high-frequency transitions, making the image appear crisper.

So,

$$\text{Sharpened Image} = \text{Figure 2} + c \times \text{Figure 3}$$

;(c = scaling factor
determining the strength
of the sharpening)

e) Yes, figure 4 can be used to sharpen figure 2 by Unsharp masking.

Steps:

1) ~~Blur the original image~~

Subtract the blurred version (figure 4) from the original clean version (figure 2). The result of this subtraction is the "mask" which is containing the fine details/edges.

$$\text{Mask} = \text{Figure 2} - \text{Figure 4}$$

2) Add this mask back to the original image (figure 2)

$$\text{Sharpened Image} = \text{Figure 2} + k \times \text{Mask}$$

; $k = 1$ $\rightarrow$ unsharp masking

; $k > 1$ $\rightarrow$ high boost filtering.

So, this increases the contrast specifically at the edges, resulting in a sharpened image.

[Lecture - 6]

[Ans to the question no 1]

Image 1

$$f(x,y) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(Complex Numbers)

$$\begin{bmatrix} 1+j & 0+j & 0+j \\ 0+j & 0+j & 0+j \\ 0+j & 0+j & 0+j \end{bmatrix}$$

magnitude spectrum:

$$|F(u,v)| = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This image represents a zero-frequency (DC) signal. It relates to "flat" area in spatial processing.

It is the result of applying an averaging (low-pass) filter where all high frequency variations are removed.

Image 2

$$g(x,y) = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

(complex numbers)

$$F(u,v) = \begin{bmatrix} 0 & -4.5+j2.6 & +5-j2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$u=0: \text{sum} = -1+0+1=0$$

$$u=1: (-1)e^0 + 0 + (1)e^{-j2\pi(2)(1)/3} = -1 + e^{-j4\pi/3}$$

$$e^{-j4\pi/3} = \cos(240^\circ) - j\sin(240^\circ) = -0.5 + j0.866$$

$$F(0,1) = -1 - 0.5 + j0.866 = -1.5 + j0.866$$

$$u=2: (-1)e^0 + (0) + (1)e^{-j2\pi(2)(2)/3} = -1 + e^{-j8\pi/3} = -1 + e^{-j2\pi/3}$$

$$e^{-j2\pi/3} = -0.5 - j0.866$$

$$F(0,2) = -1.5 - j0.866$$

$$|F(u,v)| = \begin{bmatrix} 0 & 5.2 & 5.2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$1.3 \approx 0.598$$

This image contains high-frequency changes in the horizontal direction. It relates to edge detection techniques in spatial processing (specifically vertical edge detection filter like prewitt or sobel)

[Ans to the question no 2]

(A) shows a circularly symmetric pattern, which matches the symmetry of the spatial image

[Ans to the question no 3]

Ans: (B)

The larger ring corresponds to the higher frequency of the concentric circles,

[Ans to the question no 4]

Ans: (B)

This shows an X-shaped pattern, aligning with the diagonal frequency concentrations expected from the checker board.

[Ans to the question no 5]

Image 1: The dominant feature is the horizontal line. The bottom DFT shows a distinct, strong vertical line of energy through the center, (ocean)

Image 2: This image contains repeating stripes, while somewhat chaotic, the stripes are generally vertical / diagonal. (zebras)

The top DFT shows a more diffuse distribution of energy with some diagonal "X" shapes, characteristic of complex textured patterns like the zebra stripes.

[Ans to the question no 6]

Output Image 1: [High pass Filter]

The output image shows edges and fine details while smooth, constant backgrounds (sky, grass) has turned gray/dark.

A High pass filter works by attenuating the low-frequency components and passing high-frequency components.



High pass filter

Output Image 2: [Low pass filter]

The output image is blurred or smoothed. The sharp edges of the coat and tripod are gone, and fine details are washed out.

A low pass filter works by passing the low-frequency components and attenuating the high frequency components.

